Software Project Management
Assignment-12
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following correctly match the CMMI flavor with its focus area?

a. CMMI for Development → Software and systems engineering activities.
b. CMMI for Development → Exclusive focus on supplier contracts.
c. CMMI for Services → Limited only to IT help desks.
d. CMMI for Services → Organizations providing services.
e. CMMI for Acquisition → Acquiring products and services.

Correct Answer: a. CMMI for Development → Software and systems engineering activities
d. CMMI for Services → Organizations providing services.
e. CMMI for Acquisition → Acquiring products and services

Detailed Solution:

CMMI has distinct models: CMMI for Development covers software, systems, and hardware engineering (a); CMMI for Services applies to any service-providing organization, not just IT help desks (d); and CMMI for Acquisition focuses on acquiring products and services (e). It does not exclusively focus on supplier contracts (b—that's part of acquisition, not development), and it is not limited to IT help desks (c).

QUESTION 2:

Which of the following statements about the Personal Software Process (PSP) are true?

a. PSP recognizes that an individual process differs from a team project process.
b. PSP provides a defined process for individual use.
c. PSP is designed exclusively for large teams.
d. PSP can be fine-tuned by individuals.
e. PSP eliminates the need for any development process.

Correct Answer: a. PSP recognizes that an individual process differs from a team project process.
b. PSP provides a defined process for individual use.
d. PSP can be fine-tuned by individuals.

Detailed Solution:

The Personal Software Process (PSP) is a disciplined framework designed for individual developers to plan, track, and improve their own work. It acknowledges the difference between individual and team processes (a), provides a defined personal process (b), and allows individuals to tailor it to their needs (d). It is not designed for large teams (c—that's the Team Software Process, TSP), nor does it eliminate the need for other development processes (e).

QUESTION 3:

Which of the following statements about PSP evolution are true?

a. Measurement activities appear before planning activities.
b. Quality management is introduced before personal planning.
c. Personal process evolution is a PSP3 activity.
d. Design and code reviews are introduced in PSP0.
e. PSP consists of four levels: PSP0 through PSP3.

Correct Answer: a. Measurement activities appear before planning activities.
c. Personal process evolution is a PSP3 activity.
e. PSP consists of four levels: PSP0 through PSP3

Detailed Solution:

PSP evolves through four levels: PSP0 (basic measurement of time and defects), PSP1 (adds size estimation and planning), PSP2 (introduces design and code reviews for quality management), and PSP3 (focuses on personal process evolution and scaling). Thus, measurement (a) comes before planning, personal process evolution (c) is a PSP3 activity, and PSP has four levels (e). Quality management is introduced in PSP2, after planning (b), and reviews are not in PSP0 (d).

QUESTION 4:

Which of the following statements about Six Sigma are true?

a. Six Sigma allows up to 340 defects per million opportunities.
b. A Six Sigma process produces no more than 3.4 defects per million opportunities.
c. A Five Sigma process produces about 230 defects per million opportunities.
d. A Four Sigma process produces about 6210 defects per million opportunities.
e. Four Sigma produces fewer defects than Five Sigma.

Correct Answer: b. A Six Sigma process produces no more than 3.4 defects per million opportunities.
c. A Five Sigma process produces about 230 defects per million opportunities.
d. A Four Sigma process produces about 6210 defects per million opportunities

Detailed Solution:

Six Sigma is a quality management methodology that aims to reduce defects to 3.4 per million opportunities (DPMO), making (b) correct. A Five Sigma process corresponds to about 230 DPMO (c), and Four Sigma corresponds to about 6,210 DPMO (d). Option (a) is incorrect because 340 DPMO lies between Five Sigma and Six Sigma, not at Six Sigma. Option (e) is false because Four Sigma produces more defects (6,210) than Five Sigma (230).

QUESTION 5:

Which statements explain why software reliability can vary among users?

a. Reliability is unaffected by how software is used.
b. Users may exercise different features of the software.
c. Different usage patterns can expose different failures.
d. Every user interacts with software in exactly the same way.
e. Failures visible to one user may not appear for another.

Correct Answer: b. Users may exercise different features of the software.
c. Different usage patterns can expose different failures.
e. Failures visible to one user may not appear for another.

Detailed Solution:

Software reliability can vary among users because different users may exercise different features (b), follow different usage patterns that expose distinct failures (c), and encounter failures that are visible to some but not others (e). Reliability is strongly affected by usage (a is false), and users do not interact identically (d is false).

QUESTION 6:

Which of the following statements correctly compare software and hardware failure curves?

a. Software and hardware have identical failure behavior over time.
b. Hardware typically exhibits a "bathtub curve" failure pattern.
c. Software does not wear out physically with age like hardware.
d. Hardware failure rate remains constant throughout its lifetime.
e. Software changes can introduce new failures.

Correct Answer: b. Hardware typically exhibits a "bathtub curve" failure pattern.
c. Software does not wear out physically with age like hardware.
e. Software changes can introduce new failures

Detailed Solution:

Hardware follows a "bathtub curve"—early failures, constant random failures during useful life, and increasing wear-out failures as it ages (b). Software, however, does not physically wear out (c); it can degrade due to changes, which may introduce new defects (e). Hardware failure rates are not constant throughout (d—they increase during burn-in and wear-out phases). Software and hardware do not have identical failure behavior (a).

QUESTION 7:

A POFOD value of 0.001 implies that:

a. Approximately 1 out of 1000 service requests may result in failure.
b. Approximately 10 out of 1000 service requests may result in failure.
c. The probability of failure on a demand is 0.1%.
d. The system fails once every 1000 hours.
e. The metric explicitly depends on time.

Correct Answer: a. Approximately 1 out of 1000 service requests may result in failure..
c. The probability of failure on a demand is 0.1%.

Detailed Solution:

POFOD (Probability of Failure on Demand) measures the likelihood that a system will fail when a service is requested. A value of 0.001 means there is a 0.1% chance of failure per demand, which translates to approximately 1 failure out of 1,000 service requests (a and c). It does not imply 10 out of 1,000 (b—that would be 0.01), nor is it time-based (d and e—that would describe MTBF, not POFOD).

QUESTION 8:

Which of the following statements about a reliability growth model are true?

a. It is used only after software deployment.
b. It models how software reliability grows over time.
c. Reliability improves as errors are detected and repaired.
d. It assumes reliability decreases whenever defects are fixed.
e. It can help estimate future reliability levels.

Correct Answer: b. It models how software reliability grows over time.
c. Reliability improves as errors are detected and repaired
e. It can help estimate future reliability levels

Detailed Solution:

A reliability growth model tracks how software reliability improves over time as defects are detected and fixed, making (b) and (c) correct. It is primarily used during system testing, not only after deployment (a is false). It assumes reliability increases, not decreases, when defects are fixed (d is false). It can also help estimate future reliability levels (e).

QUESTION 9:

Which of the following are implications of the Littlewood and Verall model?

a. Every bug contributes identically to reliability improvement.
b. The order of bug removal has no effect on reliability growth.
c. Reliability growth is not necessarily uniform across bug fixes.
d. Early testing may produce substantial reliability gains.
e. Remaining bugs after extensive testing may have smaller reliability impacts.

Correct Answer: c. Reliability growth is not necessarily uniform across bug fixes.
d. Early testing may produce substantial reliability gains.
e. Remaining bugs after extensive testing may have smaller reliability impacts.

Detailed Solution:

The Littlewood–Verrall model assumes that different faults have different sizes/impacts, so reliability growth is not uniform across fixes (c). Early testing tends to find the most impactful faults, leading to substantial early gains (d), while the remaining faults after extensive testing are likely to have smaller impacts (e). The model does not assume identical contributions (a), and the order of removal does affect growth because fixing larger-impact faults earlier yields more improvement (b is false).

QUESTION 10:

Which of the following statements about statistical testing are true?

a. Statistical testing uses only randomly generated source code.
b. Test cases are selected based on the expected operational profile of the software.
c. It ignores user behavior patterns.
d. Statistical testing aims to reflect actual system usage.
e. It can be used to assess software reliability.

Correct Answer: b. Test cases are selected based on the expected operational profile of the software.
d. Statistical testing aims to reflect actual system usage.
e. It can be used to assess software reliability.

Detailed Solution:

Statistical testing (also called operational profile testing) selects test cases based on the expected operational profile (b) to reflect actual system usage (d). By measuring failures during such testing, it helps assess software reliability (e). It does not use randomly generated source code (a—it uses the actual software), and it explicitly incorporates user behavior patterns, not ignore them (c is false).

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